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Abstract: Existing visual-based fatigue detection methods usually monitor drivers' fatigue by capturing their facial features, including eyelid movements, yawn frequency and head pose. However, these approaches typically do not take drivers' biological signals into consideration. An accurate model for fatigue detection requires combining both facial behavior and biological data. This paper proposes a novel non-intrusive method for driver multimodal fusion fatigue detection by extracting eyelid features and heart rate signals from the RGB video. The multimodal feature fusion method could significantly increase the accuracy of fatigue detection. Specifically, we established two fatigue detection models based on heart rate and the PERCLOS value respectively with one-dimensional Convolutional Neural Network (1D CNN), where the PERCLOS refers to the percentage of eyelid closure over the pupil. Finally, the outputs of the two models are weighted to achieve the multimodal fusion fatigue detection. Simulation results show that our method yield better performance than traditional methods.

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